MILITARY SPECIFICATION SHEET

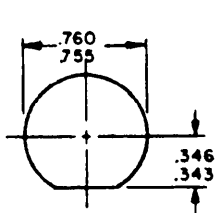
CONNECTORS, COAXIAL, RADIO FREQUENCY,

SERIES QNC (UNCABLED - RECEPTACLE, SOCKET CONTACT, HERMETIC SEALED,

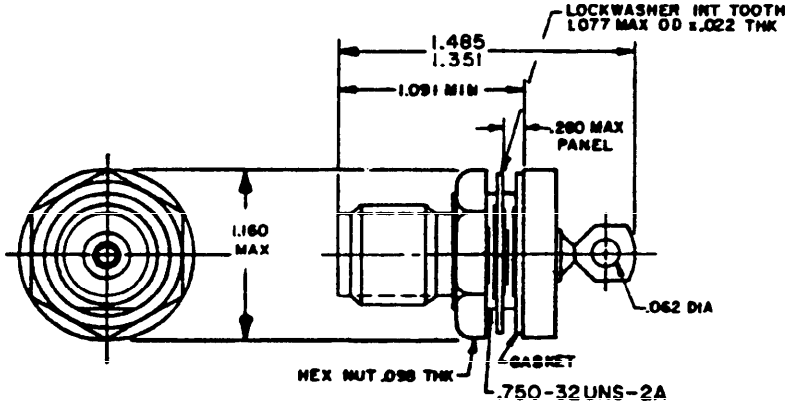
JAM NUT, FRONT AND REAR MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-39012.

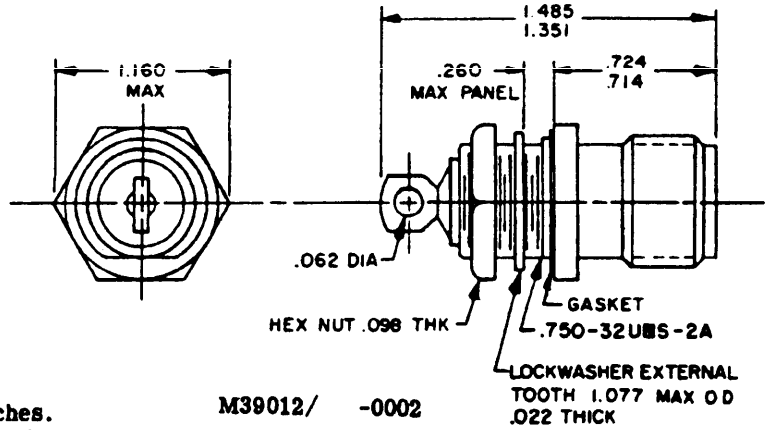


M39012/91-0001
M39012/91-0002



M39012/ -0001

INCHES	MM
.022	.56
.059	1.50
.062	1.57
.098	2.50
.260	6.60
.343	8.71
.346	8.79
.714	18.14
.724	18.39
.750	19.05
.755	19.18
.760	19.30
1.077	27.36
1.091	27.71
1.160	29.46
1.351	34.32
1.485	37.72

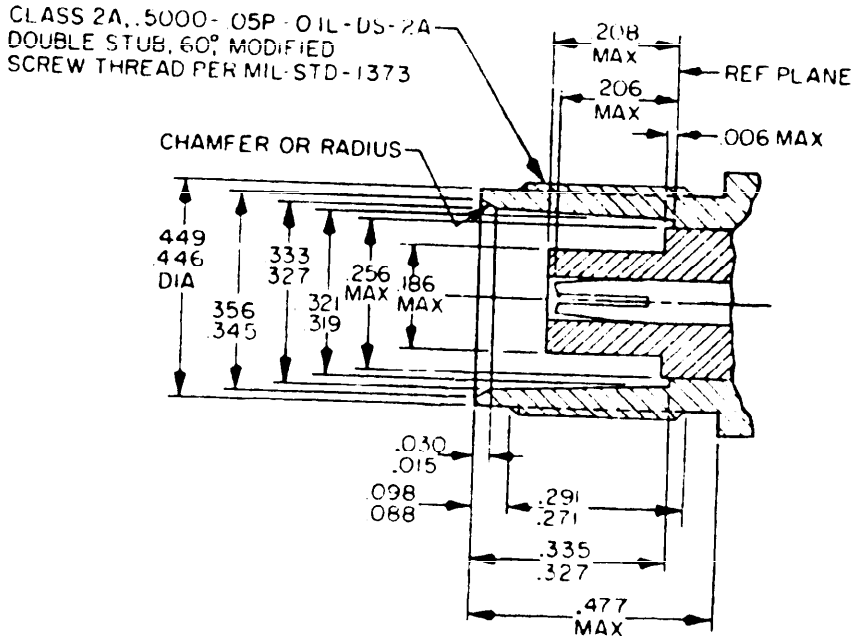


M39012/ -0002

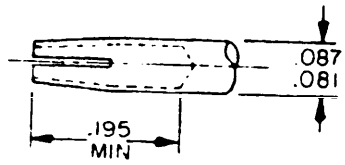
NOTES:
1. Dimensions are in inches.
2. Dimension 1.160 is the largest overall diameter.
3. Dimension 1.485 is the largest overall length.
4. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
5. Wrench flats to accommodate standard wrench opening per H-28, app. 10.

FIGURE 1. General configuration.

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INCHES	MM	INCHES	MM	INCHES	MM
.006	.15	.186	4.72	.327	8.31
.015	.38	.195	4.95	.333	8.46
.030	.76	.206	5.23	.335	8.51
.052	1.32	.208	5.28	.345	8.76
.054	1.37	.256	6.50	.356	9.06
.081	2.06	.271	6.88	.446	11.32
.087	2.21	.291	7.39	.449	11.40
.088	2.24	.319	8.10	.477	12.12
.098	2.50	.321	8.15	.5000	12.70



NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. The .256 dimension applies to that portion (if applicable) of the dielectric which protrudes beyond the metal shoulder (or reference plane) by dimension .006.

FIGURE 2. Mating dimensions for socket terminations.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating: 500 volts rms maximum working voltage. 125 volts rms maximum at 70,000 feet.

Temperature rating: -65° to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:
Longitudinal force - Not applicable.
Torque - 2 inch-pounds maximum.

Coupling proof torque:
20 inch-pounds minimum.

Inspection conditions: 4 to 6 inch-pounds.

Mating characteristics:
See figure 2 for dimensions.
Center contact (socket):
Oversize test pin - .057 diameter minimum (nonclosed entry contacts only).
Insertion depth - .125 minimum.
Number of insertions - One.

Insertion force test - Steel test pin dia .054 minimum.
Test pin finish - 16 microinches.
Insertion force - 2 pounds maximum.

Withdrawal force test: Steel test pin dia .052 maximum.
Withdrawal force - 2 ounces minimum.
Test pin finish - 16 microinches.

Hermetic seal: Leakage shall not exceed $1x10^{-5}$ cc/sec of tracer gas at atmospheric pressure.

Leakage (pressurized connectors):
Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 5,000 megohms minimum.

Center contact retention:
6 pounds minimum axial force.

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR):
Not applicable.

Connector durability:
500 cycles minimum at 12 cycles/min maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact	4.0	5.0
Outer contact	.2	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 2,000 volts rms at sea level.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Shock: Method 213, test condition I, MIL-STD-202.

Temperature cycling: Method 102, test condition C, MIL-STD-202.

Thermal shock: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity,

Corona level: Not applicable.

RF high potential withstanding voltage:
Voltage and frequency - 1,000 volts rms at 5 MHz.
Leakage current - Not applicable.

Cable retention force:
Not applicable.

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Coupling mechanism retention force:
Not applicable.

RF leakage:
Not applicable.

Insertion loss: Not applicable.

Qualification not required.

Part number: M39012/91 -0001 for the front- and M39012/91 -0002 for the rear-mounted connector.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, MI, EL
DSA - ES

User activities:
Army - AT, AV, ME

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5935-1862-9)

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